

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

I Eddu Suchier/192524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *E. Suchier*

Annexure A
CONSTRAINT-BASED PROBLEM SOLVING

3. A corporate network is experiencing repeated malware infections that spread rapidly without user interaction. Propose a containment and prevention strategy.

Constraints:

- A. The network consists of 500+ interconnected systems
- B. Downtime per system should not exceed 10 minutes
- C. Detection must occur within 2 minutes of infection
- D. Solution must differentiate between virus, worm, and Trojan behavior

The corporate network contains **500+ interconnected systems** and is experiencing malware infections that spread rapidly without user interaction. The main objective is to detect infections within **2 minutes**, contain them with **less than 10 minutes of downtime**, and distinguish between **viruses, worms, and Trojans**.

1. Proposed Containment and Prevention Strategy

The organization should deploy a layered security architecture consisting of Endpoint Detection and Response (EDR), IDS/IPS, network segmentation, automated isolation, and centralized security monitoring.

Constraint	Proposed Solution
500+ systems	Use centralized EDR and IDS/IPS to monitor all endpoints and network traffic.
Detection within 2 minutes	Use real-time behavioral monitoring and automated alerts.
Downtime ≤ 10 minutes	Automatically isolate only infected systems instead of shutting down the network.

Constraint	Proposed Solution
Differentiate malware	Analyze file, process, and network behavior to identify viruses, worms, and Trojans.
Prevent future infections	Apply security patches, firewalls, application controls, least-privilege access, and email/web filtering.

2. Malware Differentiation

- **Virus:** Attaches itself to legitimate files and spreads when the infected file is executed. Detection focuses on suspicious file modifications and replication.
- **Worm:** Automatically spreads from one system to another without user interaction. Rapid network scanning and repeated connection attempts are strong indicators.
- **Trojan:** Disguises itself as legitimate software. Detection focuses on suspicious processes, unauthorized communication, and persistence activities.

3. Containment Workflow

Detect → Classify → Isolate → Remediate → Recover

1. **Detection:** EDR agents and IDS/IPS continuously monitor the 500+ systems and network traffic. Suspicious activity generates an immediate alert.
2. **Classification:** The system analyzes the observed behavior to determine whether it is a virus, worm, or Trojan.
3. **Isolation:** The infected machine is automatically moved to a quarantine VLAN or its network access is restricted. This prevents a worm from spreading to other systems.
4. **Remediation:** Security tools terminate malicious processes, quarantine infected files, remove malware, and apply required patches.

5. **Recovery:** After confirming that the system is clean, it is returned to the normal network. Automated procedures help keep downtime within the **10-minute limit**.

4. Constraint Validation

The strategy satisfies all requirements:

- **500+ systems:** Centralized monitoring provides scalable protection.
- **2-minute detection:** Continuous real-time monitoring enables rapid identification.
- **10-minute downtime:** Automated isolation limits disruption to affected systems.
- **Malware differentiation:** Behavioral analysis identifies the different characteristics of viruses, worms, and Trojans.
- **Prevention:** Segmentation, patch management, access control, and security filtering reduce the chance of future infections.

Conclusion

The recommended solution is a layered and automated malware defense system using EDR, IDS/IPS, network segmentation, and automated quarantine. The approach follows detect → classify → isolate → remediate → recover, allowing the organization to contain rapidly spreading malware while maintaining network availability and satisfying all given constraints.

4. An organization needs to deploy a firewall architecture to protect its internal network from external threats and insider attacks. Design the solution.

Constraints:

- A. Must support both packet filtering and application-level inspection
- B. Latency introduced should not exceed 5 ms per request
- C. Should handle at least 1 Gbps traffic
- D. Must allow secure remote access for employees

The organization needs a firewall architecture that protects the internal network from external threats and insider attacks while maintaining high performance. The solution must support packet filtering, application-level inspection, low latency, 1 Gbps throughput, and secure remote access.

1. Proposed Firewall Architecture

A Next-Generation Firewall (NGFW) should be deployed at the network perimeter, combined with internal segmentation firewalls and a VPN gateway.

Internet → Edge Router → NGFW → DMZ → Internal Firewall → Internal Network

- **NGFW:** Performs packet filtering, application-level inspection, intrusion prevention, and traffic control.
- **DMZ:** Hosts public-facing services such as web and mail servers separately from the internal network.
- **Internal Firewall:** Controls traffic between departments and sensitive systems, helping prevent insider attacks.
- **VPN Gateway:** Provides encrypted remote access for authorized employees.

2. Constraint Handling

Constraint	Proposed Solution
Packet filtering + application inspection	NGFW performs IP/port-based filtering and Layer-7 application inspection.
Latency ≤ 5 ms	Use hardware-accelerated firewall processing and optimized security rules.

Constraint	Proposed Solution
Traffic $\geq$ 1 Gbps	Select a firewall rated for more than 1 Gbps throughput under the required security features.
Secure remote access	Provide SSL/IPsec VPN with strong authentication and access controls.

3. Security Measures

The firewall should use stateful packet inspection, application-aware policies, intrusion prevention, malware detection, and URL filtering. Access should follow the least-privilege principle, allowing users and applications only the traffic they require.

For remote employees, an IPsec or SSL VPN should be used with multi-factor authentication (MFA). VPN users should receive access only to the resources required for their job rather than unrestricted access to the internal network.

4. Constraint Validation

- **Security:** External traffic is filtered at the perimeter, while internal segmentation limits insider threats.
- **Performance:** Hardware acceleration and optimized policies help keep latency below 5 ms per request.
- **Capacity:** A firewall with sufficient headroom can handle the required 1 Gbps traffic.
- **Remote access:** VPN with MFA provides secure encrypted connections for employees working remotely.

Conclusion

The recommended architecture is a layered NGFW-based solution with a DMZ, internal segmentation firewall, and secure VPN access. It provides both packet-level and application-level protection while meeting the 5 ms latency, 1 Gbps throughput, and secure remote-access constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided